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Analysis of Carbon Quantum Dot Antifouling Coating on an Electrochemical
Biosensor for the Multiplex Detection of Breast Cancer Biomarkers

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Introduction

The current gold standard of mammography for breast cancer diagnosis has inherent risks and disadvantages such as expensive equipment, inaccessibility, inability to distinguish benign/malignant tumours, and sensitivity to dense tissue more prevalent in younger patients. This has led to a 30% misdiagnosis rate in breast cancer detection [1]. The need for a simple and quick diagnosis method is apparent. There are a multitude of biomarkers attributed to breast cancer that, if detected, can ensure a timely diagnosis and reduce patient mortality [2]. A multiplexing solution capable of analyzing a sample for a panel of cancer-associated biomarkers is essential as the expression levels of these biomarkers (higher on average in cancer patients) can overlap with those found in healthy individuals [3].

Electrochemical (EC) biosensors have proven their ability to recognize analytes in small sample sizes with low detection limits in relatively quick amounts of time [4]. Addressing the shortcomings of mammography, EC biosensors can provide quick, cost-effective, accurate diagnosis of malignant tumours through the multiplex detection of relevant biomarkers. However, before EC biosensors can reliably perform in point-of-care (POC) settings, the issue of fouling – the adsorption of nonspecific substances on an interface, must be addressed. This is a major issue in the field of EC biosensors that can lead to reduced sensitivity or the inhibition of signal generation in less than a minute [4], [5], [6]. Extensive research on antifouling coatings using nanomaterials to enhance conductive properties is key to addressing this problem. There are several iterations of coating methods that have utilised gold nanoparticles (AuNPs), gold nanowires (AuNWs), carbon nanotubes (CNTs), reduced graphene oxide (rGO), and functionalized reduced graphene oxide (prGOx) [4], [5], [7], [8]. Furthermore, several studies have independently developed their own derivatives of 3-dimensional (3D) nanocomposite coatings for antifouling [9], [10], [11]. Carbon quantum dots (CQDs) have received extensive attention for their antifouling properties in wastewater treatment and a recent study was one of the first to propose research expanding their antifouling properties against oxidative products in the detection of insulin [6], [12]. The potential for CQDs as a cost-effective and efficient replacement of nanomaterials in current EC antifouling coatings should be investigated.

Consequently, this research proposes an EC biosensor for the detection of miRNA-21 and miRNA-1246, breast cancer biomarkers found within exosomes in whole blood [13], [14], [15]. This project proposes an antifouling coating consisting of CQDs, bovine serum albumin (BSA), and glutaraldehyde (GA) to address the shortcomings of current isolation/purifying techniques for exosomes in sample preparation [16]. The electrochemical biosensor platform will use hairpin probe (HPP) aptamer bioreceptors that work in conjunction with the redox mediators to generate a signal. See the Methodology section for the EC biosensor mechanism of action.

This project will answer the following: (1) Are CQDs a viable nanomaterial for antifouling purposes in EC biosensors and how do they compare with other nanomaterials? (2) How do aptamer bioreceptors compare to immune bioreceptors in their capacity for multiplexing, based on experimental results and literature benchmarks?

Research Hypothesis and Objectives

The design objectives for this project are organized into a five-year timeline with the following long-term objectives (short-term objectives are placed beneath their respective long-term objectives):

Design CQD coating for initial validation of antifouling and conductivity properties. (2 yrs)

- Matrix evaluation for optimal concentrations of CQD, BSA, and GA in nanocomposite.

- Develop CQD/BSA/GA composite protocol with fabrication times under 16 hours.
- Adapt method for deposition of antifouling coating on electrode surface (~1-5 minutes).

Validate HPP design for selectivity, sensitivity, and functionality within EC Environment. (3 yrs)

- Validate selectivity to specific target sequence (3:1 target-to-mismatch signal ratio)
- Test functionality in complex environments (human serum, whole blood) against control.
- Determine sensitivity with calibration curve and dynamic range. Compare with literature.

Demonstrate effectiveness and stability of CQDs in antifouling coating for EC detection in complex environments. (1 yrs)

- Signal reduction less than 5% when exposed to complex environments for 30 minutes.
- Comparison of antifouling properties with other nanomaterial performance in literature.
- Signal reduction less than 10% when exposed to complex environment for 1 month.

Design of the CQD coating must occur during the first two years of the project before the evaluation of the selectivity, sensitivity, and functionality of the HPP design. However, design of the coating and HPP can overlap to ensure both are complete before the end of the third year. Demonstrations of coating effectiveness and long-term stability can be tested within the last year. Preparations for publishing a research paper can also start in the fourth year of the project. The specific values for the short-term goals are set to achieve comparable values with existing literature. The total budget is estimated at \$465,250 with a team involving one PhD student and two MASc students.

Literature Review

Antifouling Coatings Literature Review

Research has extensively investigated antifouling coating, involving methods such as polyethylene glycol (PEG), self-assembled monolayers (SAM), and zwitterionic polymers. However, these coatings can suffer from instability through degradation or irregular formation leading to a reduction in electrode sensitivity [4], [9], [10], [17]. A review of antifouling strategies in EC biosensors by Russo et al. (2021) covers the latest developments in antifouling coatings [17]. This review will cover coatings using nanomaterials within a 3D matrix that are relevant to this research proposal.

A polyaniline nanowire (PANI-NW) BSA/GA antifouling coating method was proposed by Li et al. (2021) on glassy carbon electrodes (CEs). The paper tested their method through the detection of immunoglobulin G using peptide aptamers with a reported LOD of 0.27 ng/mL [9]. The coating demonstrated good conductivity with a signal reduction of only 8.4% for the whole apparatus when exposed to undiluted whole blood for 30 minutes [9]. However, long-term stability testing under undiluted human serum saw a ~20% signal reduction at 10 days. It was noted that the fabrication of the PANI-NW/BSA/GA requires an incubation time of 16 hours [9]. Further research is required to optimize the antifouling properties of this method to ensure better performance, especially including whole blood testing which incorporates other foulants such as fibrinogen and coagulation factors that are not present in human serum. For example, an alternative study on supercapacitor electrodes found the addition of CQDs in a PANI/carbon fiber matrix improved capacitance and stability [18].

Another method proposed an immunosensor using gold-coated silver nanowires in a BSA/GA matrix on screen-printed CEs for the detection of APO-A1 – a protein overexpressed by patients with bladder cancer. This method achieved an LOD of 22 pg/mL and a dynamic range of 100 pg/mL to 250 ng/mL [11]. The sensor experienced a ~12% current density reduction after a month-long exposure to human serum and urine [11]. While the nanowire design reduces costs by

substituting a gold wire for a thin gold coating, further research should examine other materials that may require less complex fabrication with comparable properties.

Li et al. (2021) introduced a method utilising multi-walled carbon nanotubes (MWCNTs) in an electrochemical biosensor for the purpose of detecting HbA1c [5], [10]. This method successfully detected HbA1c with a reported LOD of 0.4% (i.e., 0.8 ug/mL HbA1c in 199.2 ug/mL HbA0) with a dynamic range of 2 to 15%. Additionally, this coating method demonstrated the ability to maintain a signal at 88% current density after one month of exposure to unprocessed human serum [10]. Further optimizations on this method should investigate reducing fabrication protocol time (~18 hours) for deposition of coating on electrodes [10].

Carbon nanotubes are affected by signal decay if subjected to successive signal measurements which may limit its stability [4]. Additionally, the selection of nanomaterials used may inhibit the method's scalability and cost, with more complex fabrication processes reducing the coatings' applicability to large-scale production. Another group addressed these concerns by building on their previous research and developing a coating consisting of reduced graphene oxide with amine-terminated groups, BSA, and crosslinking GA [4], [5], [7], [8]. This method successfully demonstrated its antifouling and conductive ability by maintaining ~90% of its current density in unprocessed whole blood and achieved a LOD of 64.5 pg/mL for the detection of procalcitonin [8]. All this is achieved with a dip coating fabrication method that takes ~1 minute, improving previous iterations of the design by ~24 hours [5], [8]. However, a limitation of dip coating methods is their lack of precision in deposition. This may affect EC biosensors that rely on screen-printed electrodes as the reference and control electrode surfaces tend to be in close proximity to the working electrode. This may lead to interference in signal generation.

Multiplexing Literature Review

A recent review by O'Brien et al. (2024) covers the latest research on EC biosensors for the multiplexed detection of breast cancer biomarkers, emphasizing the use of CEs with the incorporation of nanotechnology [2]. The following will examine EC biosensors for the detection of miRNA-21 and miRNA-1246 that are relevant to this research proposal.

Pothipor et al. (2022) proposed a dual-electrode system using an antibody (anti-CA 15-3) and DNA (miRNA-21) capture probe to detect breast cancer [19]. This method is built on a nanocomposite base composed of plastic and 2D materials (i.e., poly(3-aminobenzylamine), molybdenum selenide, and graphene oxide). Gold nanoparticles functionalized with their respective capture probes and unique redox mediators are deposited on the coated electrode surface (i.e., 2,3-diaminophenazine and toluidine blue). These unique NPs generate a distinguishable signal when their respective target analyte binds to the probe. The paper reported an LOD of 0.14 U/mL and 1.2 fM for the antibody and DNA probe, respectively [19]. However, this multiplexing method is limited by the number of nanoparticles with unique redox characteristics that can provide a distinct signal. The addition of more biomarkers reduces this method's scalability as each new biomarker requires a unique redox mediator, increasing the complexity of the sensor fabrication method.

Another study proposed the simultaneous detection of miRNA-155, miRNA-21, and miRNA-16 using silver-gold nanoparticles (AgAuNPs) labeled with capture DNA bioreceptors and Cd, Pb, and Cu, respectively. The target was allowed to bind to their corresponding AgAuNPs in solution phase, creating a DNA/RNA hybrid. Consequently, the carbon electrodes coated in S9.6Ab – an antibody that targets DNA/miRNA hybrids, captured the AgAuNPs to generate a signal. This resulted in a reported LOD of 0.98 fM, 3.58 fM, and 0.25 fM for miRNA-155, miRNA-21, and miRNA-16, respectively [20]. However, it was noted that sensor stability was only predictably

ensured for 14 days and the nanoparticles with Pb observed a signal reduction of 20.8% after 28 days, indicating that some metals may have less stability than others for long-term storage. This limitation will hinder the selection of potential metal candidates for other target analytes. Additionally, this sensor fabrication method also suffers from the same limitations as the biosensor proposed by Pothipor et al. (2022) in that additional multiplexing also increases the fabrication complexity of the entire biosensor.

Conclusion

A solution that offers the flexibility of detecting multiple biomarkers while minimizing the addition of complex fabrication steps for each biomarker present is required. Avoiding signal generation through unique selection of redox mediators or the selection of metals that exhibit distinct signals will eliminate this issue. Additionally, there has been extensive research on different nanomaterials utilised in antifouling coatings for EC biosensors. Adding to that literature, the investigation of CQDs has the potential to provide another nanomaterial that is applicable.

Methodology

The proposed EC biosensor in a point-of-care setting works by applying a drop of sample, prepared from a commercial prep kit (ex. ExoCIB – exosome isolation kit) on the working electrode. The surface of the electrode is protected by an antifouling coating composed of CQD/BSA/GA and functionalized with HPPs through amine linkers at the 3' end. When the target binds to the HPP, its structure will change from a loop to a linear duplex, exposing biotin at the 5' end. Consequently, HRP-streptavidin is applied to the surface and allowed to bind to the exposed biotin. A washing step occurs to remove unbound HRP and TMB precipitate is applied to the surface. Measurements are taken to determine the signal generated from the local precipitation of TMB, ensuring multiplexing capabilities. If the target analyte is not present, the HPP structure stays in its loop form, preventing HRP binding to the electrode surface. The HRP is washed off and no signal is generated. The following outlines protocols/experiments for coating and sensor fabrication. Certain steps may be subjected to optimization or further tests to determine exact timings/concentrations.

CQD Fabrication and Validation (PhD)

CQD Fabrication Protocol: Dissolve candle soot-derived carbon black in DMF and sonicate (cost-effective and simple base material) [6], [21]. Centrifuge solution to separate by density before collecting supernatant fluid. Dry fluid at high temperature to achieve powdered CQDs.

CQD Characterization: TEM for CQD size distribution. EIS for conductivity properties.

Antifoul Coating Fabrication and Deposition (MAsc #1, PhD)

Antifouling Coating Protocol: Mix concentrations of CQDs and BSA in PBS to form nanocomposite. Sonicate the solution for uniform dispersion of CQDs. Centrifuge mixture to remove aggregates and recover supernatant fluid. Enable crosslinking by adding GA.

Deposition Method: Clean gold electrodes before drop-casting nanocomposite on working electrode. Allow incubation (~1 minute) before performing washing step in PBS. Quench unreacted GA with ethanolamine. Apply washing step in PBS.

HPP Surface Functionalization: Activate coating surface with EDC/NHS – EDC is a carbodiimide reagent that activates carboxyl groups to form an intermediate which react with amines and is stabilized by NHS. Incubate HPP probes to enable functionalization. Apply washing step.

HPP Fabrication (MASC #2, PhD)

HPP SELEX: SELEX used for design of loop portion of HPP. The HPP probe will be modified with 5' biotin and 3' amine functional groups.

HPP Optimization: Optimize the concentration of HPP to ensure biotin is not accidentally exposed on electrode surface or to prevent overcrowding of HPP on surface.

Surface Plasmon Resonance: Assess selectivity and binding kinetics.

Antifouling Coating Properties and Validation (MASC #1, PhD)

Characterization of Coating: SEM/AFM for pore size, film thickness, and surface roughness. XPS for verification of chemical groups on surface. CV for current density measurements.

Optimization of Material Concentrations: Test differing concentrations of CDQ (0.5-15 mg/mL) BSA (0.5-10 mg/mL), and GA. Incubate 1% BSA and human serum for 30 minutes. Determine best concentration for maintaining current density (CV).

Long-term Stability Tests: Expose sensors to 1% BSA and human serum for a one-month period. Measure current density reduction and compare with control.

Storage Tests: Assess sensor performance after storage of three months at room temperature, 4 °C, and -20 °C. Test in 1% BSA and human serum to determine current density reduction.

HPP and Biosensor Validation (MASC #2, PhD)

Selectivity Test: Compare sensor response to target sequence, sequence with one mismatch, sequence with two mismatches, and noncognate sequence. Measure results in current density.

Functionality Test: Examine sensor ability to perform in complex environments. Test in PBS, human serum, and whole blood samples. Compare performance with literature results.

Sensitivity Test: Determine calibration curve, LOD, and dynamic range of sensor under non-ideal environments (ideal if sensor is not working well). Compare performance with literature results.

Alternative Methods (PhD)

Examining the AuNP/BSA/GA coating proposed by Sabaté del Río et al. (2019), that did not work because the NPs were larger than the pore size of their 3D matrix, is an alternative coating method that still allows the comparison of nanomaterials [5]. Additionally, a PEG-based self-assembled monolayer is a common approach that can be done simultaneously with the CQD/BSA/GA coating for a comparative study against conventional antifouling methods [17]. Furthermore, to address the second research question stated in the introduction, a comparative benchmark study using ELISA to determine the efficiency of aptamer-based biosensors against antibody-based biosensors should be addressed.

Impact

The social impacts of this research will demonstrate the viability of CQDs in a biomedical application, potentially shifting perceptions and building confidence in their application within the biosensing field. In contrast to their metallic counterparts, prior research has demonstrated that carbon nanoparticles have comparably low cytotoxicity levels (cell viability ~90% in trypan blue and PI uptake assay at 500 ug/mL, <12% toxicity in red blood cells at 150 ug/mL) and contribute minimal oxidative stress damage [22], [23], [24], [25]. CQDs have already demonstrated their applicability in antifouling coatings for wastewater treatment and further research will evaluate their potential in other applications [12]. Additionally, the economic impact of this research is felt through the promotion of adopting CQDs as a cost-effective alternative that is comparable to other current nanomaterials. As a whole, the biosensor contributes towards a quick and accessible method of diagnosing breast cancer while mitigating the need for expensive equipment [1].

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